

SAMPLE EXAM QUESTIONS for Water Infrastructure

- (a) Restoration of waterways is becoming increasingly important to ensure ecological and social benefits.
- List the three primary functions of Reference Sites.
 - It is important to know what to measure or monitor for a successful restoration of a stream. List five potential evaluation tools and criteria for each of the following objectives:
 - Improve aquatic habitat
 - Improve riparian habitat
- (b) Lane's Qualitative Geomorphic Model is often used for alluvial channel design to determine changes to riverbed aggradation or degradation. Describe what would happen in an alluvial channel that is currently in balance between degradation and aggradation under these situations (indicated changes in the balance to aggradation or degradation):
- An irrigation take is placed upstream
 - A landslide occurs upstream resulting in high sediment yields
 - Large size gravel is extracted upstream resulting in finer sediment downstream
- (c) Poorly designed hydraulic structures can fail, but proper planning and design consideration help reduce the likelihood of failure.
- Describe the two main hydraulic causes for failure in hydraulic structures.
 - Characterizing the hydrology is an important step in a detailed hydraulic structure design. Describe two ways to obtain your design flows.
 - List two factors you should account for which could result in significant future hydrological changes.
- (d) You have been tasked to design a chute. A key aspect of the design is to determine the appropriate length (L) of the stilling basin at the bottom of the chute. The design criteria for the chute and stilling basin suggest using the maximum value of either $L = 3 * d_2$ or $L = 3 * D_2$.

Where,

d_2 is the subcritical depth after an expected hydraulic jump

$D_2 = 1.5$ m (the normal depth of the channel downstream)

The flow through the chute is $1.5 \text{ m}^3/\text{s}/\text{m}$. The slope of the chute is 0.33 thus the energy loss is specified from relevant Figure in the supplementary material for this test. Assume the chute has 9 meters of drop from the upstream energy level to the toe of the chute. You may need to solve equations in this problem iteratively by "guess and check" or "trial and error".

- (e) A small town near the Waimakariri River has been granted a consent to divert $6.08 \text{ m}^3/\text{s}$ of water from the river to supply the town 8 km away. The natural slope of the terrain is 0.005 and the channel is to be built through a colloidal alluvial silt soil. The water sourced from the river is transporting colloidal silts. Assume the roughness in the channel bed will be $n = 0.025$ and that you are trying to keep the peak water depth in the channel at or below 0.5 m. Assume the channel will be rectangular (sides lined) and the width will be 8 m.
- Calculate a channel design slope.
 - How many drop structures (and what height) would you recommend to minimize costs?
- (f) Three parallel culverts under a road in Rolleston need to be replaced to account for an increase in flow in a drainage channel. There are only two sizes available: 1050 mm or 900 mm diameter culverts. The client wants to know which size they should buy to meet the following conditions:
- Max $Q = 3 \text{ m}^3/\text{s}$ (total combined flow for the 3 culverts)
 - h_o (depth of tail water in channel downstream) = 1 m (assume the pipe flows full)
 - Allowable head water (H) is 1.0 m
 - $k_e = 0.5$
 - Length = 100 m
 - Slope of culvert = 0.01 m/m
 - $n = 0.012$
 - The entrance of the culvert has a bevelled lip of 45 deg.

Use Figures provided in the back of the test to check the feasibility of your proposed culvert diameters. Show your work and clearly indicated which values were obtained from figures.

For rectangular structures:

$$Q=VA \text{ or } q=yV$$

$$Fr = \frac{V}{\sqrt{gy}}$$

$$E = y + \frac{V^2}{2g}$$

For a trapezoidal channel:

The critical shear stress for the sides/bottom is expressed as follows:

$$\tau_{cs} = \left[1 - \frac{\sin^2 \alpha}{\sin^2 \phi} \right]^{1/2} \cdot \tau_{cb}$$

Hydraulic jump:

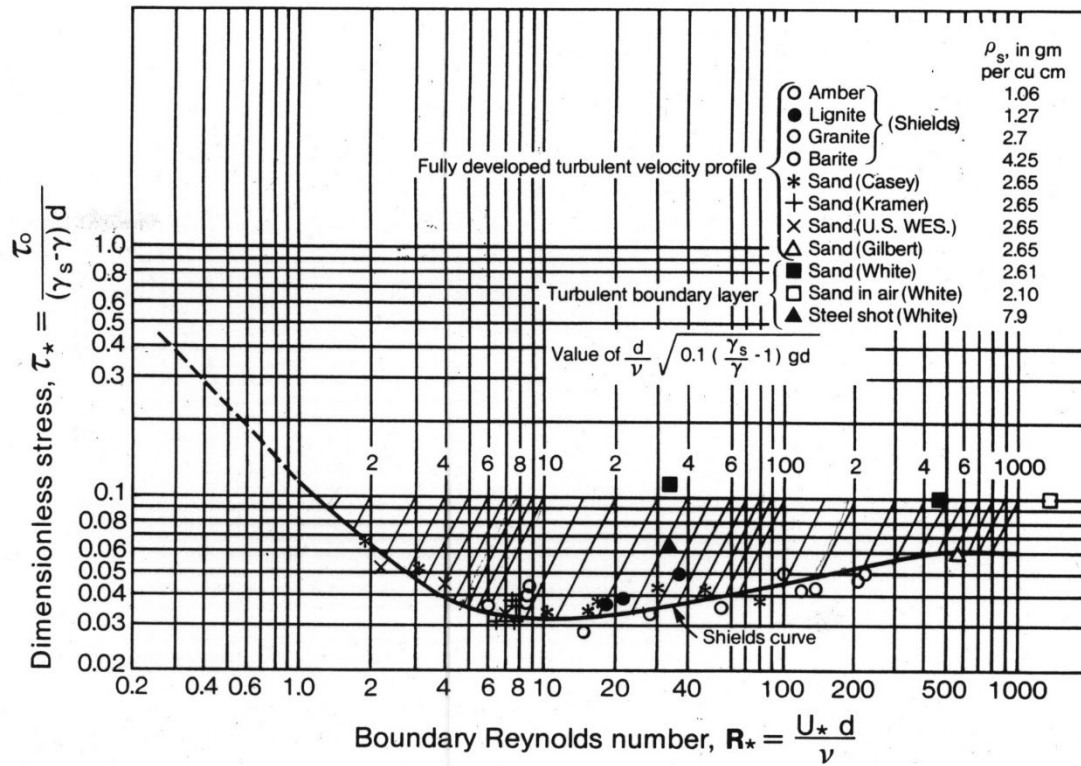
$$\frac{y_2}{y_1} = \frac{1}{2} \left(\left(\sqrt{1 + 8Fr_1^2} \right) - 1 \right)$$

Manning's formula:

$$V = \frac{1}{n} R^{2/3} S^{1/2}$$

Table 1. Allowable velocities for different channel materials.

Original material excavated for canals	Mean velocity, for straight canals of small slope, after aging with flow depths less than 3 ft (0.9 m)					
	Clear water, no detritus		Water transporting colloidal silts		Water transporting noncolloidal silts, sands, gravels, or rock fragments	
	ft/s	m/s	ft/s	m/s	ft/s	m/s
Fine sand (noncolloidal)	1.5	0.46	2.5	0.76	1.5	0.46
Sandy loam (noncolloidal)	1.75	0.53	2.5	0.76	2.0	0.61
Silt loam (noncolloidal)	2.0	0.61	3.0	0.91	2.0	0.61
Alluvial silt (noncolloidal)	2.0	0.61	3.5	1.07	2.0	0.61
Ordinary firm loam	2.5	0.76	3.5	1.07	2.25	0.69
Volcanic ash	2.5	0.76	3.5	1.07	2.0	0.61
Stiff clay (very colloidal)	3.75	1.14	5.0	1.52	3.0	0.91
Alluvial silt (colloidal)	3.75	1.14	5.0	1.52	3.0	0.91
Shales and hardpans	6.0	1.83	6.0	1.83	5.0	1.52
Fine gravel	2.5	0.76	5.0	1.52	3.75	1.14
Graded, loam to cobbles (when noncolloidal)	3.75	1.14	5.0	1.52	5.0	1.52
Graded silt to cobbles (when colloidal)	4.0	1.22	5.5	1.68	5.0	1.52
Coarse gravel (noncolloidal)	4.0	1.22	6.0	1.83	6.5	1.98
Cobbles and shingles	5.0	1.52	5.5	1.68	6.5	1.98



Note: "d" in meters

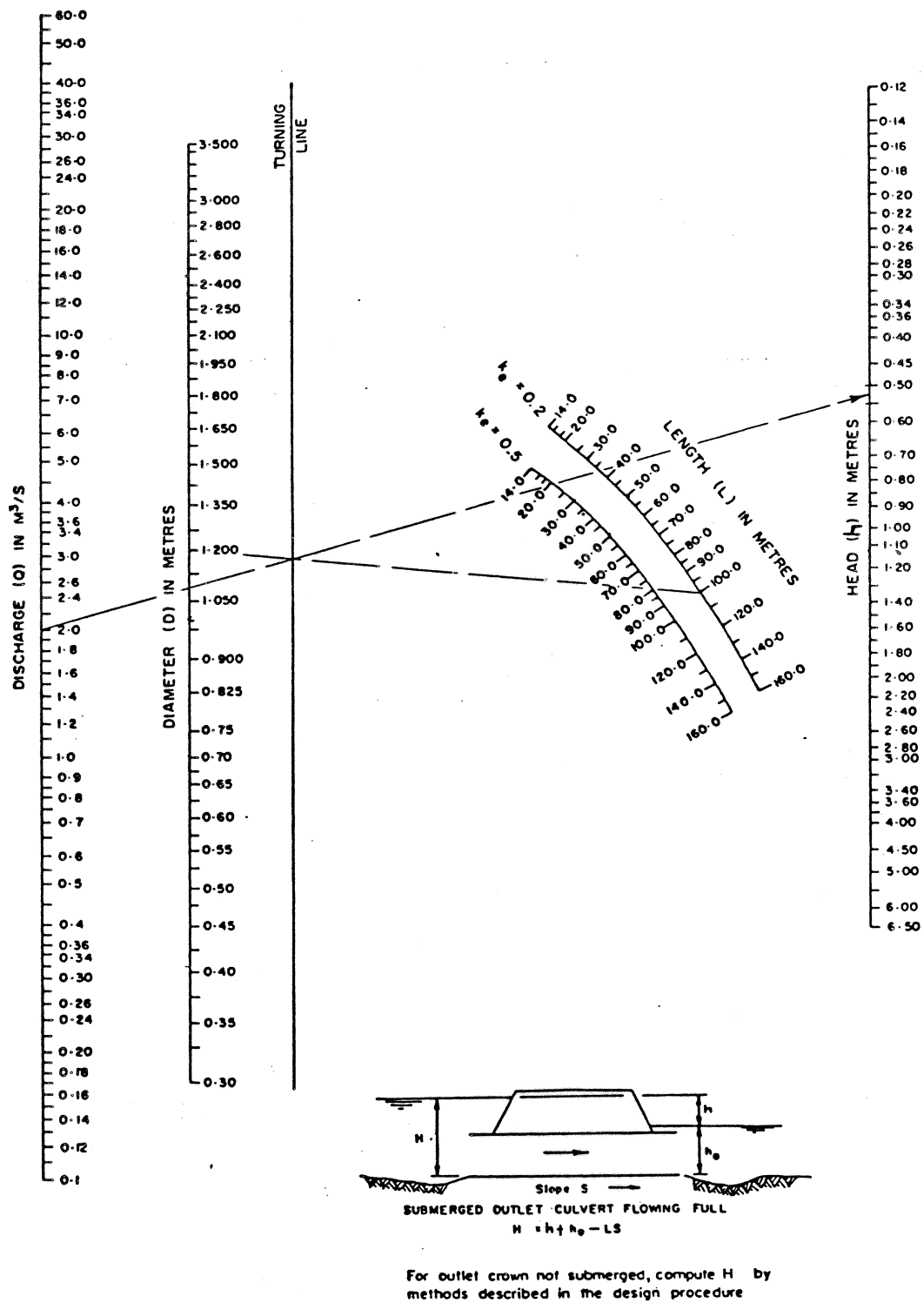
Figure 1. Shields shear stress diagram.

The dimensionless stress for the Shields curve can be approximated using the following equation:

$$\tau_* = 0.22 \beta + 0.06 \times 10^{-7.7\beta}$$

$$\beta = \left(\frac{1}{\nu} \sqrt{\left(\frac{\gamma_s - \gamma}{\gamma} \right) g d^3} \right)^{-0.6}$$

Note: "d" in meters, $g = 9.8 \text{ m/s}^2$



CONCRETE PIPE CULVERTS FLOWING FULL
 $n = 0.012$

Figure 2. Nomograph for head loss h , under Outlet Control – Circular Culvert.

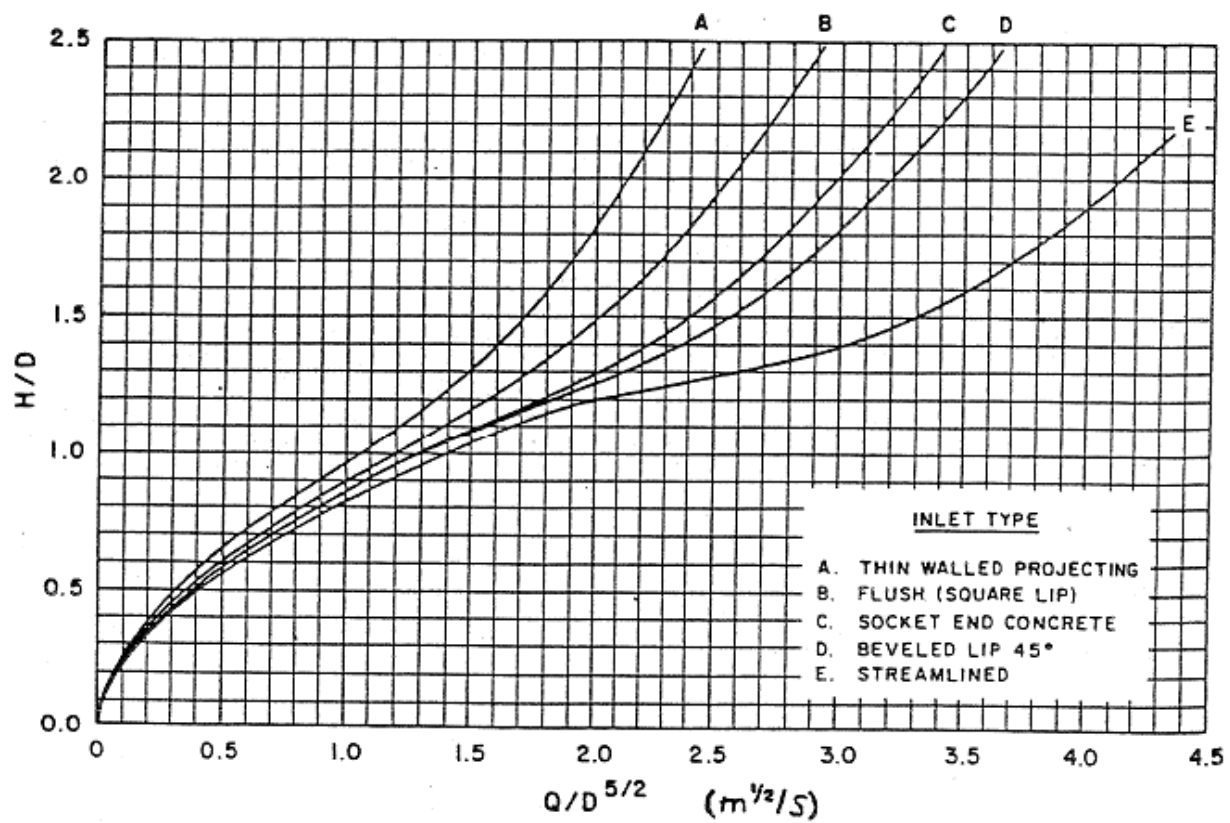


Figure 3. Rating curves for Inlet Controlled flow through culverts of diameter D.

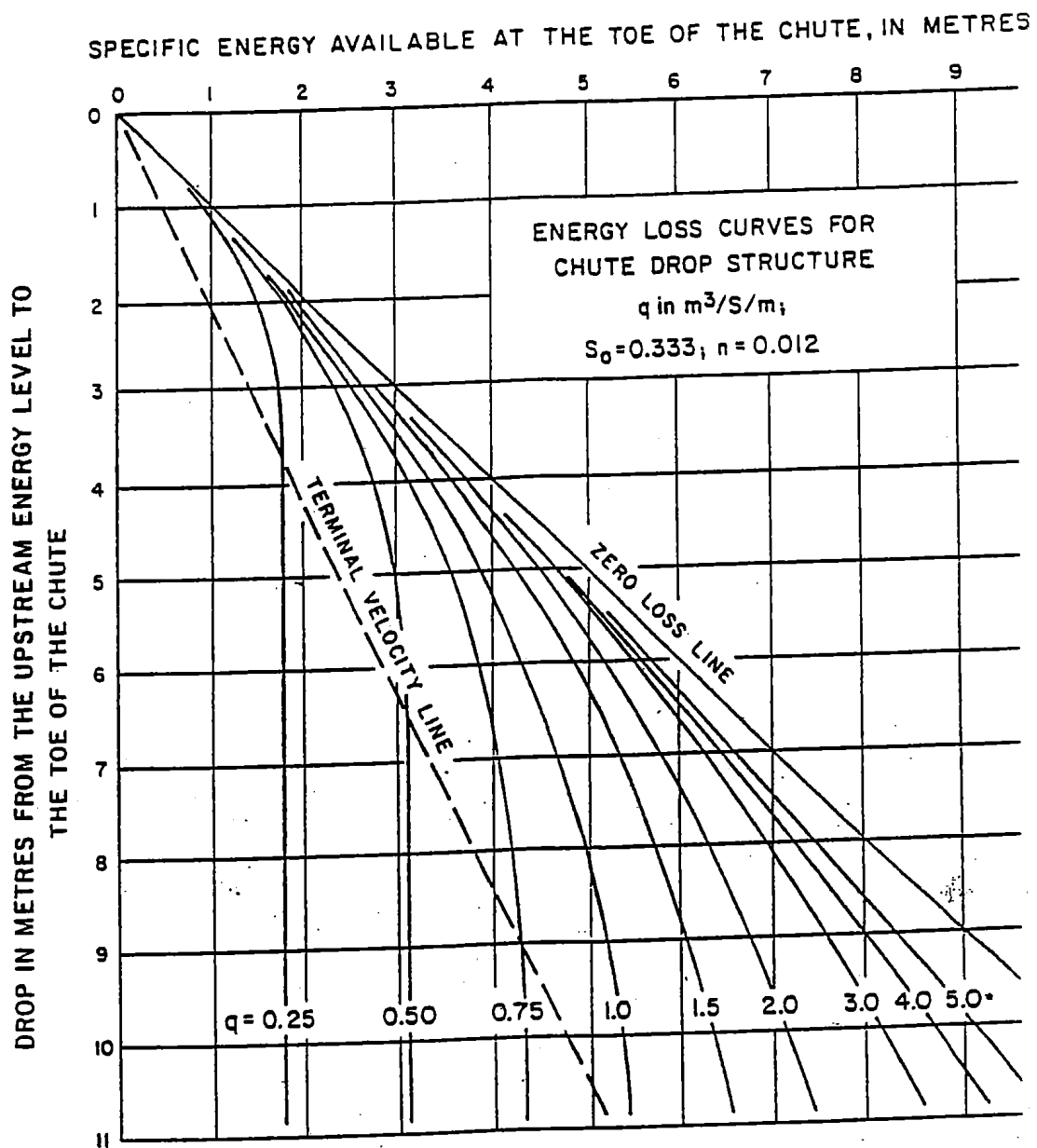


Figure 4. Energy loss curves for chute drop structure.